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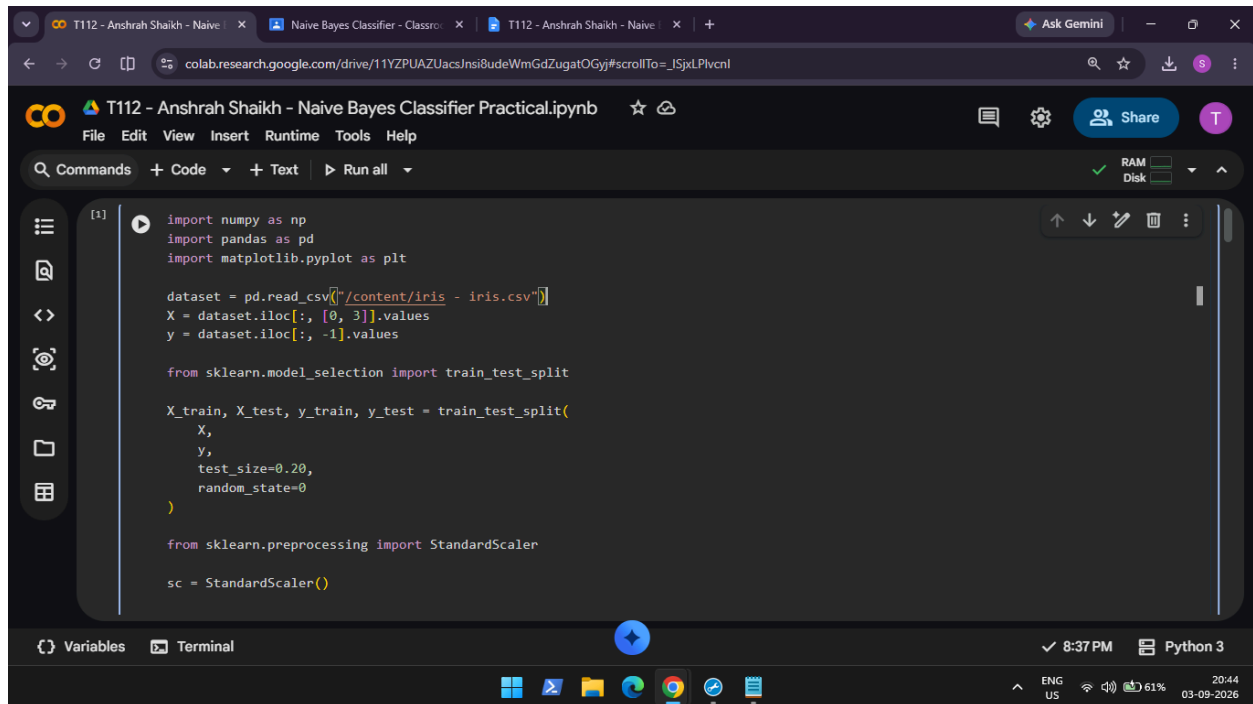
Aim: Implement the Naive Bayes' algorithm for classification.

Train a Naive Bayes' model using a given dataset and calculate class probabilities.

Evaluate the accuracy of the model on test data and analyze the results.

Dataset: iris.csv

Input:



```
[1] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

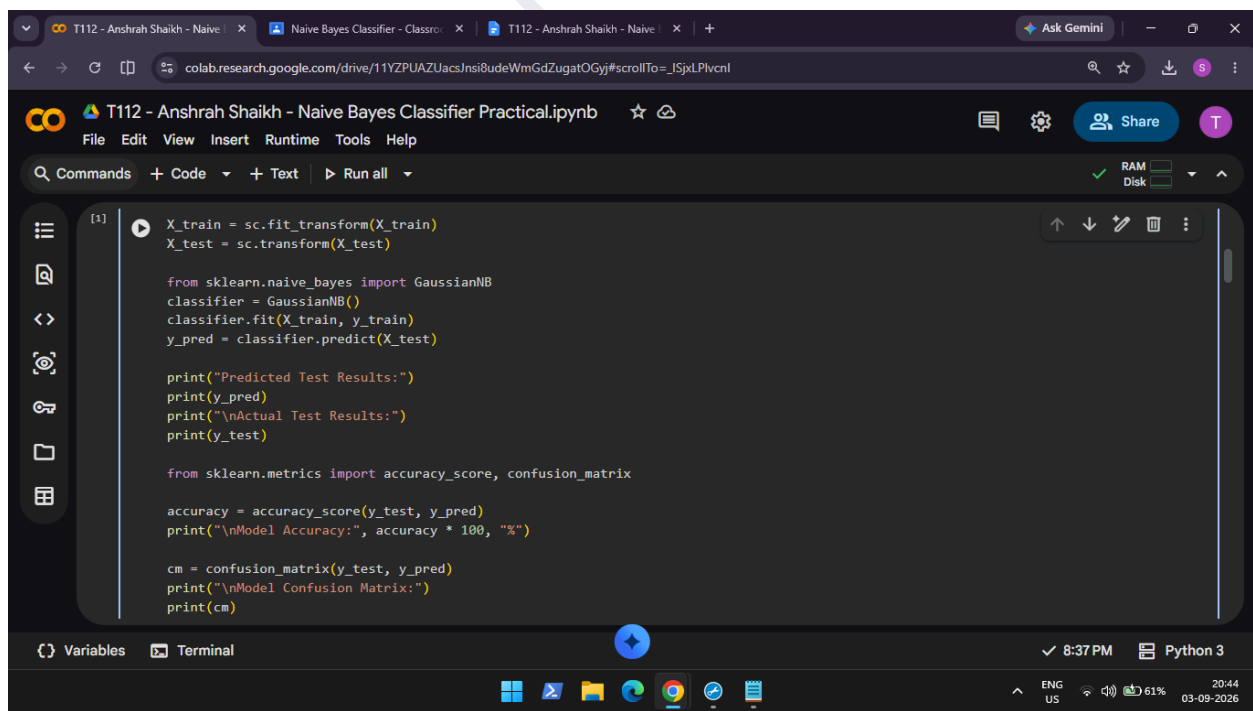
dataset = pd.read_csv("/content/iris - iris.csv")
X = dataset.iloc[:, [0, 3]].values
y = dataset.iloc[:, -1].values

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0
)

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
```



```
[1] X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)

print("Predicted Test Results:")
print(y_pred)
print("\nActual Test Results:")
print(y_test)

from sklearn.metrics import accuracy_score, confusion_matrix

accuracy = accuracy_score(y_test, y_pred)
print("\nModel Accuracy:", accuracy * 100, "%")

cm = confusion_matrix(y_test, y_pred)
print("\nModel Confusion Matrix:")
print(cm)
```

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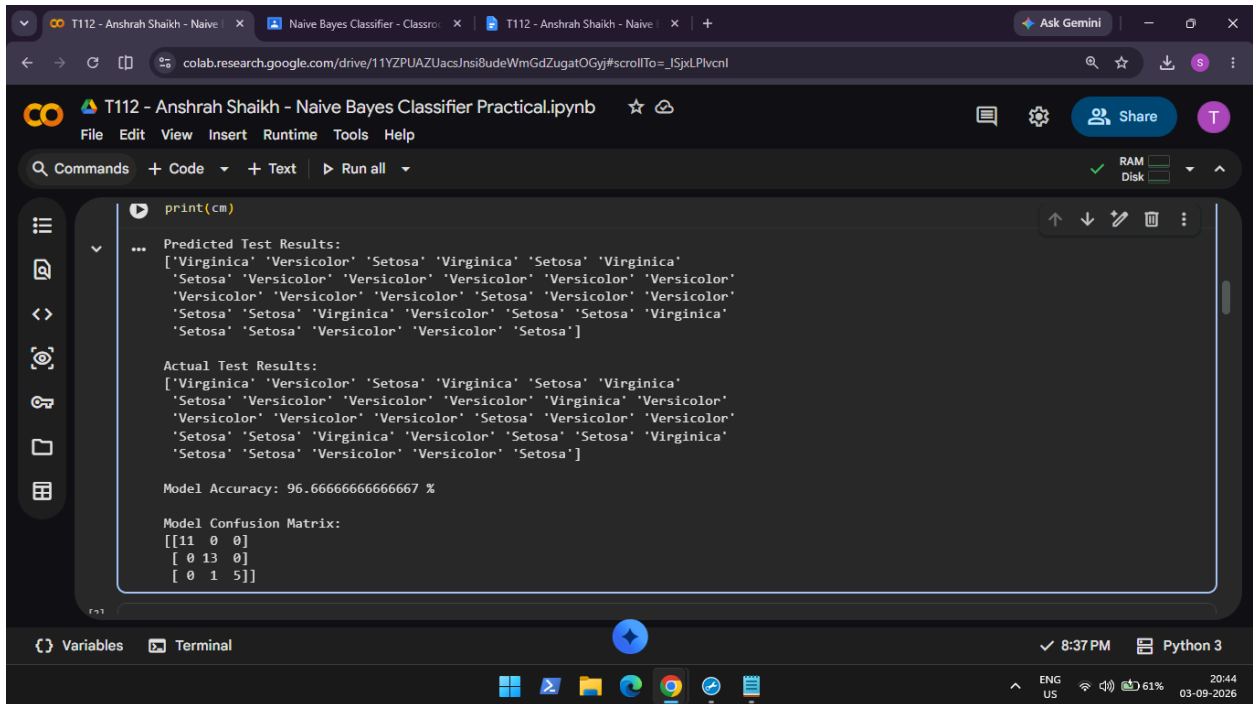
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Output:



```
print(cm)

...
Predicted Test Results:
['Virginica' 'Versicolor' 'Setosa' 'Virginica' 'Setosa' 'Virginica'
 'Setosa' 'Versicolor' 'Versicolor' 'Versicolor' 'Versicolor' 'Versicolor'
 'Versicolor' 'Versicolor' 'Versicolor' 'Setosa' 'Versicolor' 'Versicolor'
 'Setosa' 'Setosa' 'Virginica' 'Versicolor' 'Setosa' 'Setosa' 'Virginica'
 'Setosa' 'Setosa' 'Versicolor' 'Versicolor' 'Setosa']

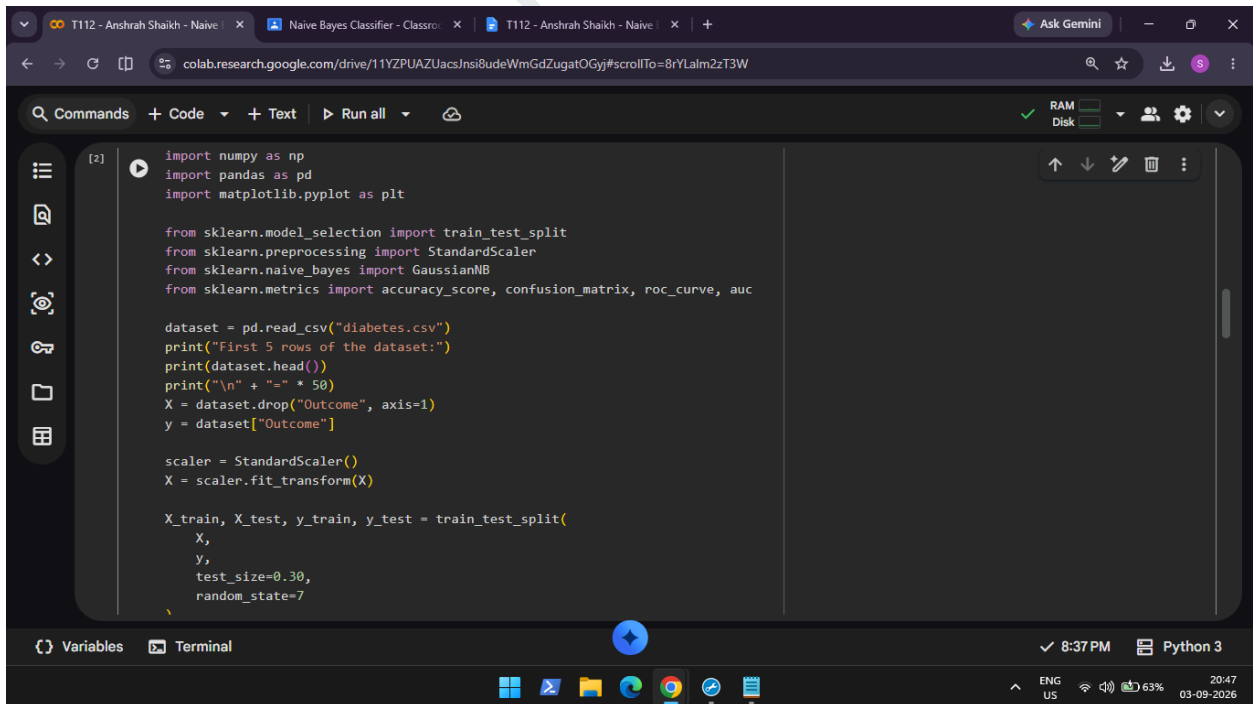
Actual Test Results:
['Virginica' 'Versicolor' 'Setosa' 'Virginica' 'Setosa' 'Virginica'
 'Setosa' 'Versicolor' 'Versicolor' 'Versicolor' 'Virginica' 'Versicolor'
 'Versicolor' 'Versicolor' 'Versicolor' 'Setosa' 'Versicolor' 'Versicolor'
 'Setosa' 'Setosa' 'Virginica' 'Versicolor' 'Setosa' 'Setosa' 'Virginica'
 'Setosa' 'Setosa' 'Versicolor' 'Versicolor' 'Setosa']

Model Accuracy: 96.66666666666667 %

Model Confusion Matrix:
[[11  0  0]
 [ 0 13  0]
 [ 0  1  5]]
```

Dataset: diabetes.csv:

Input:



```
[2] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc

dataset = pd.read_csv("diabetes.csv")
print("First 5 rows of the dataset:")
print(dataset.head())
print("\n" + "=" * 50)
X = dataset.drop("Outcome", axis=1)
y = dataset["Outcome"]

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=7
```

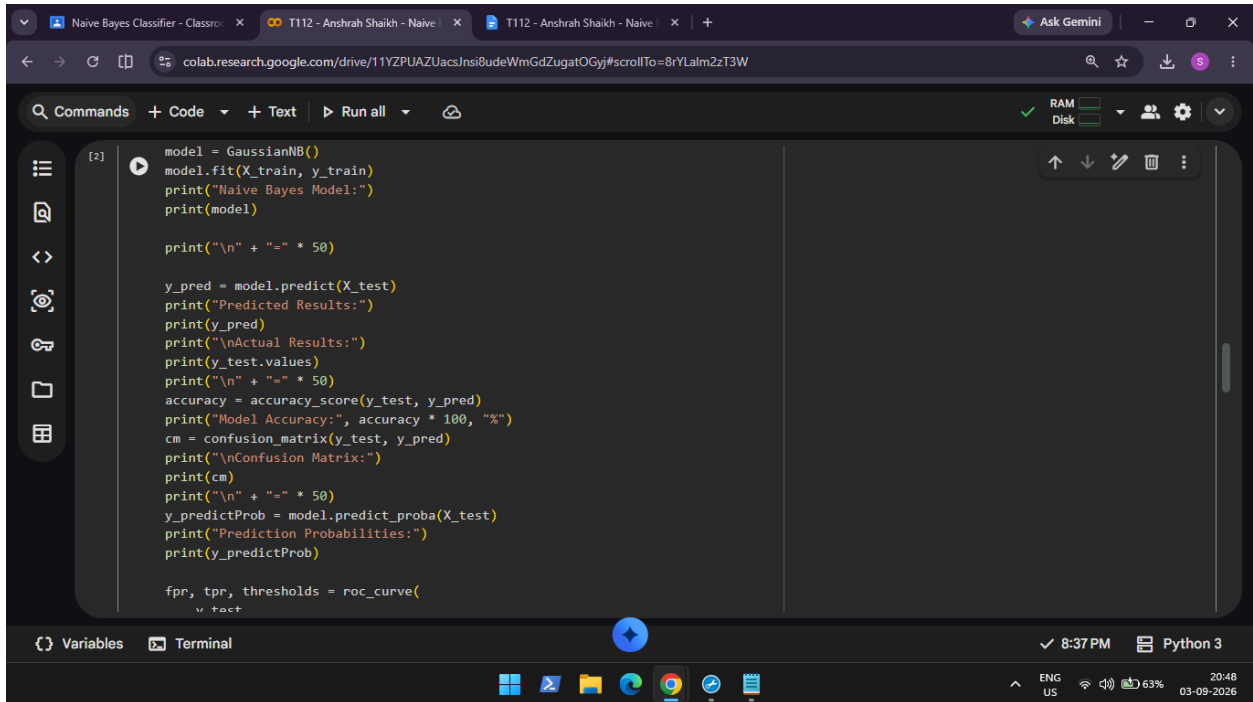
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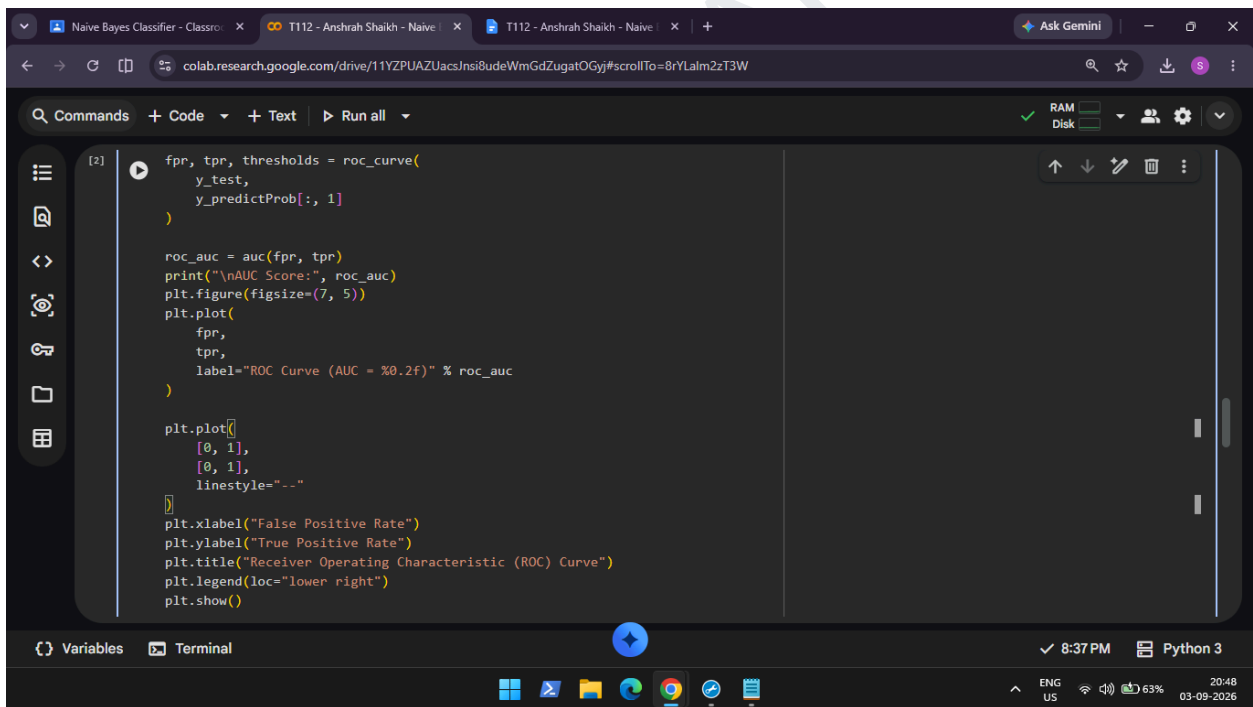


```
[2] model = GaussianNB()
model.fit(X_train, y_train)
print("Naive Bayes Model:")
print(model)

print("\n" + "-" * 50)

y_pred = model.predict(X_test)
print("Predicted Results:")
print(y_pred)
print("\nActual Results:")
print(y_test.values)
print("\n" + "-" * 50)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "-" * 50)
y_predictProb = model.predict_proba(X_test)
print("Prediction Probabilities:")
print(y_predictProb)

fpr, tpr, thresholds = roc_curve(
    y_test,
    y_predictProb[:, 1])
```



```
[2] fpr, tpr, thresholds = roc_curve(
    y_test,
    y_predictProb[:, 1])

roc_auc = auc(fpr, tpr)
print("\nAUC Score:", roc_auc)
plt.figure(figsize=(7, 5))
plt.plot(
    fpr,
    tpr,
    label="ROC Curve (AUC = %0.2f)" % roc_auc
)

plt.plot(
    [0, 1],
    [0, 1],
    linestyle="--"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Receiver Operating Characteristic (ROC) Curve")
plt.legend(loc="lower right")
plt.show()
```

Output:

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```
Naive Bayes Classifier - Classro... T112 - Anshrah Shaikh - Naive... T112 - Anshrah Shaikh - Naive... Ask Gemini
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Commands + Code + Text Run all RAM Disk

First 5 rows of the dataset:
Pregnancies Glucose BloodPressure SkinThickness Insulin BMI \
0 6 148 72 35 0 33.6
1 1 85 66 29 0 26.6
2 8 183 64 0 0 23.3
3 1 89 66 23 94 28.1
4 0 137 40 35 168 43.1

DiabetesPedigreeFunction Age Outcome
0 0.627 50 1
1 0.351 31 0
2 0.672 32 1
3 0.167 21 0
4 2.288 33 1

=====
Naive Bayes Model:
GaussianNB()

=====
Predicted Results:
[0 1 1 0 0 0 0 1 0 1 0 1 1 1 0 0 0 0 0 1 0 1 1 0 0 0 1 1 0 0 0 0 1 0 0 0
0 1 1 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 1 1 1 0 0 1 1 0 1 1 0 0
1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 1 0 0 1 0 0 1 0 0 1 0 0 1 1 0
1 0 0 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 1 0 0 1 1 0 1 0 0
1 1 0 1 0 0 0 0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 0 1 1 0 0 0 1 1 1 0 0 0 0 0
0 0 0 0 1 1 1 1 1 0 0 1 0 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0 1
1 0 0 1 0 0 1 0 0]

Actual Results:
```

```
Naive Bayes Classifier - Classro... T112 - Anshrah Shaikh - Naive... T112 - Anshrah Shaikh - Naive... Ask Gemini
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Commands + Code + Text Run all RAM Disk

Actual Results:
[0 1 1 0 1 1 0 1 0 0 1 0 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 0 0 0 1 0 0 1
0 1 1 1 1 0 0 0 0 1 1 1 1 0 0 0 0 1 0 0 0 0 1 0 0 0 1 1 1 0 1 1 1 1 1 0 0
0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 0 1 0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 0 0 0 1 0
1 1 0 1 0 0 0 1 1 0 0 0 0 0 1 0 0 0 1 0 1 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0 0
1 0 0 1 1 0 0 1 1 1 1 0 1 0 0 0 1 0 0 0 0 1 0 1 1 0 1 0 0 1 1 0 0 0 0 1
0 0 0 0 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 1 0 0 0 0 0 1 1 0 0 1 0 0 1 0 1 0 0
1 0 1 0 1 0 1 0 0]

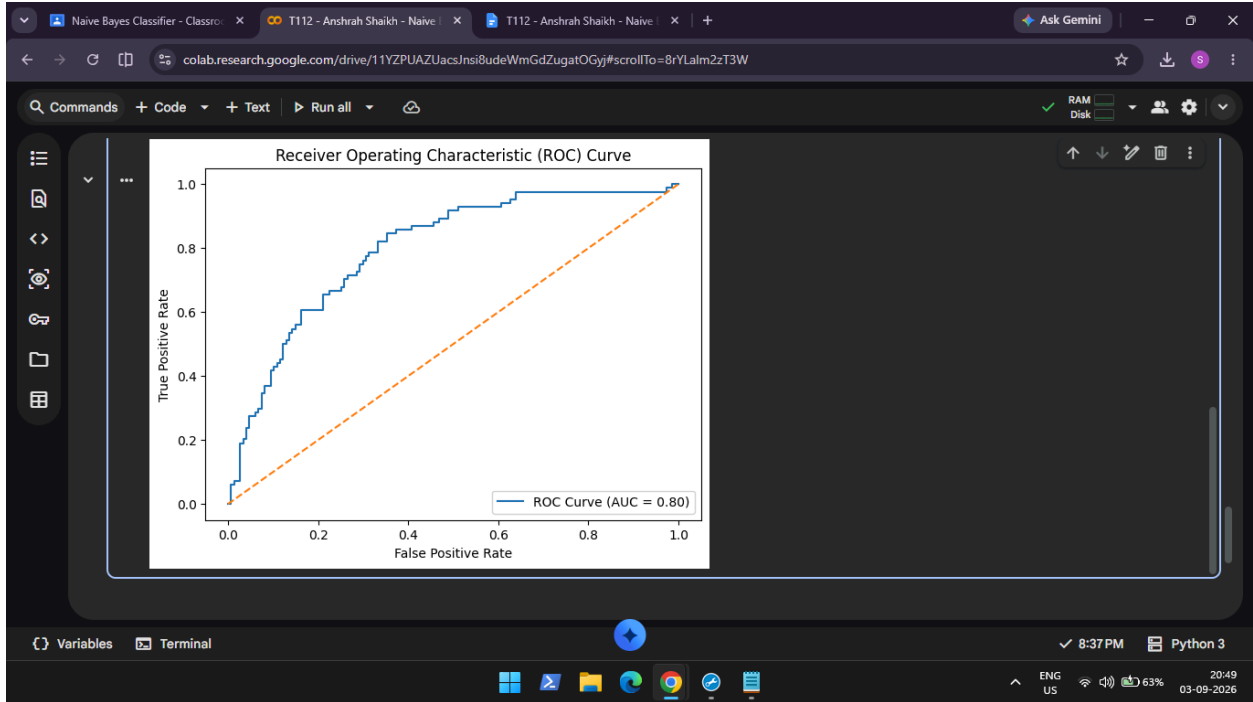
=====
Model Accuracy: 74.02597402597402 %

Confusion Matrix:
[[116  31]
 [ 29  55]]

=====
Prediction Probabilities:
[[9.86158573e-01 1.38414272e-02]
 [3.12380377e-02 9.68761962e-01]
 [5.40635227e-02 9.45936477e-01]
 [9.65065125e-01 3.49348746e-02]
 [7.50736253e-01 2.49263747e-01]
 [6.17355267e-01 3.82644733e-01]
 [9.73541945e-01 2.64580550e-02]
 [9.49535030e-01 5.04649697e-02]
 [1.97506776e-03 9.98024932e-01]
 [9.33568191e-01 6.64318089e-02]
 [3.96675687e-02 9.60332431e-01]
 [9.25723305e-01 7.42766952e-02]]
```

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